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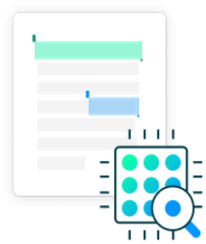

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MECHANICAL AND CHEMICAL ENGINEERING UNIT

POROUS ACTIVATED CARBON FROM WASTE PET FOR CO_2 CAPTURE

SEMESTER 2, ACADEMIC YEAR 2025–2026

Submitted in Partial Fulfilment of the Requirements for the

Bachelor's degree in chemical Engineering

Submitted by:

ID: 16S2076 NAME: HADIL HAMOOD SAID AL-HARASIYA

ID: PT12S2439198 NAME: ZAINAB MOHAMED HABIB AL LAWATI

DECLARATION

We understand that all our project work must be our own unaided work. If we make use of material from any other sources, we must clearly identify it as such in any interviews, reports, or examinations. We understand that our reports must be written unaided in our own words, apart from any quoted material which we must identify clearly in the correct manner.

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Student Name: HADIL AL-HARASIYA ID: 12J22251 Signature:

Student Name: ZAINAB AL LAWATI ID: PT12S2439198 Signature:

Supervisor Name: Dr. Ibrahim Nasser Al Siyabi

Date of submission: 12/6/2025

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This is to Certify that the Project titled "Porous Activated Carbon from Waste PET for CO_2 Capture" is the bonafide record of work done by Hadil Hamood Said Al-Harasiya and Zainab Mohamed Habib Al-Lawati, and submitted as a partial requirement for the completion of **Bachelor in Chemical Engineering** in the Semester 2 of Academic Year 2025 – 2026.

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Panel Members:

Sl. No.	**Name**	**Signature**
1.	**HADIL HAMOOD SAID AL-HARASIYA**	HADIL
2.	**ZAINAB MOHAMED HABIB AL LAWATI**	

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ABSTRACT

The two issues tackled in this project are the accumulation of plastic waste and rising concentration of carbon dioxide (CO_2) emission. In this study, a sustainable approach is suggested in which the waste plastics are converted to porous activated carbon for better CO_2 capture capability, which was achieved by chemical activation by KOH and high-temperature carbonization under an inert N2 atmosphere. This allows better development of the micropores and results in much higher surface area thus making such material suitable for adsorption of gasses. The prepared activated carbon was then treated with diethylamine to obtain further enhancement of the adsorption performance. This is change is introducing amine functional groups on the surface that will increase the amount of CO_2 which can be captured via any chemical interactions present as well as any physical adsorption. Plastic preparation, KOH impregnation, carbonization, washing, drying and amine modification are the steps of the procedure. The product material's efficiency was measured by the CO_2 adsorption test. The results indicate that waste plastics can be successfully converted to good performance activated carbon. Sustainable, low cost and eco-friendly way for CO_2 capture that simultaneously helps reduce plastic waste and CO_2 emission.

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CHAPTER I

Introduction

The need for carbon control and sustainability of the environment on a global scale has acted as a catalyst for a major shift towards creative engineering solutions. The development of successful carbon capture and storage (CCS) technology is no longer an option, but a necessity as greenhouse gas emissions, especially carbon dioxide (CO2), cause a disruption in Global Climate patterns. Plastic garbage is one of the most important environmental challenges of the 21st century – millions of tonnes of non-biodegradable polymers accumulate in ecosystems every year. This study utilizes "Waste-to-Wealth" methodology to address both of these problems optimally. In this study very porous activated carbon is produced by chemical activation of plastic waste by potassium hydroxide (KOH) followed by surface modification using diethyl amine (DEA). The resulting material is meant to be used as a high performance adsorbent that will have better chemical attraction to CO2 molecules. This chapter provides a detailed review of the study, its history, the key issues, the objectives of the study, as well as the scope and significance of the study in relation to the objectives of industry and environment of the Nation of Oman.

1.1 Background

Global warming is an issue that has become a growing concern in the environment owing to the escalation of greenhouse gas emissions, particularly the emission of carbon dioxide (CO2). Creating effective Carbon Capture and Storage (CCS) technology is crucial for "Oman Vision 2040" and the country's goal to net-zero emissions. At the same time, local ecosystems are seriously threatened by the buildup of non-biodegradable plastic garbage. By turning plastic garbage into valuable porous activated carbon, this project uses a "Waste-to-Wealth" approach that offers a sustainable route for waste management and climate change mitigation.

1.2 Statement of the problem

Commercial CO_2 capture techniques now in use frequently depend on pricey ingredients or energy-intensive procedures. Although plastic trash is a possible source of carbon, it is still technically difficult to turn it into a high-performance adsorbent with precise CO_2 selectivity. Low-cost, locally manufactured adsorbents that optimise pore shape for effective carbon sequestration through chemical activation and surface modification are few. The demand for a sustainable adsorbent made of plastic that strikes a compromise between increased chemical affinity and large surface area is addressed by this study.

1.3 Objectives

The primary goal of this research is to synthesize and evaluate porous activated carbon derived from plastic waste for CO_2 capture. The specific objectives are:

- To carbonise and chemically activate plastic trash using potassium hydroxide (KOH) in order to create activated carbon.
- To increase the activated carbon's ability to adsorb CO2 by modifying its surface with diethylamine (DEA).
- To use cutting-edge analytical methods to describe the synthesised adsorbent's chemical and physical characteristics.
- To evaluate the modified carbon's ability to adsorb CO2 in a lab setting.

1.4 Scope and Limitation of the project

This project's scope is restricted to producing activated carbon from mixed plastic waste and polyethylene terephthalate (PET). The use of KOH as the activating agent at a regulated temperature of 700 °C is the special emphasis of the chemical activation procedure. Diethylamine (DEA) is the only chemical that can be used for surface functionalisation. The study does not address the long-term industrial life cycle or large-scale pilot plant installation; instead, experimental testing will be carried out at a laboratory scale to assess the initial adsorption capability.

1.5 Significance of the project

The significance of this study lies in the fact that it provides a dual solution to the environmental issues of the Oman, a country that is hamstrung in meeting the needs of its oil and natural gas production through environmental concerns. It promotes the circular economy and lessens the burden on landfills by recycling plastic waste. In addition, achieving the decarbonisation of the Omani energy and industrial sectors is directly assisted by the development of a cost effective CO_2 adsorbent. On success this initiative can yield a template that can be applied for the manufacture of locally made high-tech environment materials from garbage.

1.6 Definition of terms

- Activated Carbon: This is a form of carbon with small low volume pores which provide a greater surface area for adsorption.
- Chemical Activation: This is a process that involves impregnation of precursor with a chemical, such as KOH, to increase the porosity in the carbonized precursor.
- Adsorption: Attachment of atoms, ions or molecules to a surface from a gas or liquid.
- Diethylamine (DEA): Secondary amine that is added as a functionalizing agent to enhance the chemical selectivity of the adsorbent to CO_2.

CHAPTER II

LITERATURE REVIEW

2.1 BACKGROUND OF CO_2 CAPTURE AND WASTE-PLASTIC UTILISATION:

In this context, the background information will be provided by 2.1 BACKGROUND OF CO_2 CAPTURE AND WASTE-PLASTIC UTILISATION:

Carbon Dioxide (CO_2) emissions are increasing because of industrialization, combustion of fossil fuels and rapid population growth thereby making the CO2 mitigation a global problem. The report of the intergovernmental panel on climate change (IPCC) indicates that large-scale use of CO2 capture, utilization and storage (CCUS) technology has to be adopted to limit global warming to 1.5 °C [22]. In recent years, the traditional absorption process with solvents has expanded to include improved solid adsorbents, ionic liquids, porous polymers, electrochemical methods and hybrid processes.

The increasing worries about plastic waste drive research efforts to convert waste polymers to valuable carbon materials at the same time. The adsorbents often extracted from waste materials have high surface area and tunable pore size, and shows strong affinity towards CO_2, which makes them as interesting candidates for sustainable CO_2 capture applications [23, 24]. The existing CO2-capture technologies are summarised in this chapter as well as the development of new sorbents, new regeneration technologies, and gaps that must be addressed to develop porous polymer foams from waste plastics.

2.2 CO_2 CAPTURE TECHNOLOGIES

The process of CO_2 capturing technologies aims to selectively remove CO_2 from industrial gas streams, CRM or ambient air. These processes can be classified into four groups: Absorption, Adsorption, membrane separation and recent electrochemical or hybrid processes. Each has its pros and cons, depending on the requirements for energy, the gas composition and operating conditions.

To date, the most advanced absorption pathways in terms of being commercial are still those that are aqueous amine based, but have high regeneration energies, corrosion and degradation of the solvents [25]. Adsorption technologies use solid materials, such as activated carbon, zeolites, MOFs and waste-derived carbon sorbents which physically or chemically bind CO2. These technologies are often associated with increased thermal stability, reduced regeneration energy, and other advantages [24]. Membrane technologies have selective permeability depending on differences between the diffusivity and/or solubility of CO2, thus requiring a relatively compact and modular building arrangement, but there are difficulties with plasticization and low flux [23].

Finally, the mild conditions of the capture/release processes using newly developed electrochemical, pH-swing and redox-active processes offer prospects for lower energy and electrically driven separation systems [26]. Together, these technologies form the basis for current methods of CO2 capture, and serve as point of reference for high-tech materials explained in later sections..

2.2.1 CO_2 CAPTURE ROUTES

CO2 collection can occur at a large number of steps throughout the industrial system ranging from direct air capture (DAC), oxy-fuel combustion, post-combustion and pre-combustion. Post combustion capture (PCC) captures CO_2 from flue gas after combusting the fuel and is the most widely applied method currently, as it is compatible with existing power plants (IEA, 2024) [23]. Pre-combustion capture separates CO_2 by converting fuel to syngas so that the molecules are easier to separate at high pressure. Oxyfuel systems are a technology that burns fuel in pure oxygen to deliver a nearly clean CO_2 exhaust stream which needs little separation (IPCC, 2021) [22]. However, high energy consumption is still an issue, so DAC systems are critical for future DOE applications of negative emission technologies (NETs), as they directly remove CO2 from the air [24] (Fu et al., 2022). These capture pathways surge in the operational context of the materials and procedures that are studied in ongoing CO2 capture research.

2.2.2 REGENERATION AND DESORPTION METHODS

Gómez and Gutiérrez [27] discuss the limitations of the other common strategies of desorption such as temperature-swing desorption and pressure-swing desorption, which often require high amounts of energy and may lead to some degradation of sorbent materials over time. The authors have outlined some novel alternatives like microwave swing adsorption (MSA), magnetic induction swing adsorption (MISA), electric swing adsorption (ESA), ultrasound swing adsorption (USSA), light induction swing adsorption (LISA) and pH-swing adsorption (pHSA), which are very efficient and promising methods. However these methods focus on reducing heat loss, on providing energy directly to the active areas of the sorbent and on increasing the speed of the regeneration cycles and the CO2 purity in these cycles when compared to the bulk heating methods. While early tests indicate improved energy efficiency and quicker carbon capture regeneration times, Gómez and Gutiérrez stress that challenges such as scaling up, material stability and the integration at a larger CO2 capture system remain significant hurdles. The paper goes on to highlight the importance of upcoming research efforts involving process optimization and techno-economic evaluation of those novel desorption technologies to determine their industrial viability.

2.2.3 COMPUTATIONAL TOOLS AND MACHINE LEARNING FOR MOF DESIGN

According to Mashhadimoslem et al. [28], in their work Computational and Machine Learning Methods for CO_2 Capture The design and optimization of metal-organic frameworks (MOFs) for effective CO2 capture have been greatly advanced by the combination of computational modelling and machine learning (ML). The authors show that using quantum mechanical and molecular force-field data to train an ML model is valid and can accurately model the adsorption behavior of MOFs, atomic interactions, and the stability of the structures in several contexts. They also reveal the ability to quickly screen large MOF collections, identify high performance, and predict CO_2 uptake potential with minimal experiment efforts. Moreover, the study highlights the growing importance of digital data repositories and text-mining tools for generating useful datasets that can be used to train models. However, Mashhadimoslem et al. [28] state that there remain challenges: Limited availability of consistent, machine-readable experimental data, and a need for better generalization of the models among different MOF families. Overall, the study demonstrates the ability to develop next-generation MOFs with greater CO2 capture efficiency more quickly when combining computational simulations with machine learning..

2.3 SOLID ADSORBENTS FOR CO_2 CAPTURE

2.3.1 GENERAL SOLID ADSORBENTS

2.3.1.1 ADSORPTION MECHANISMS

Patel et al. [29] provides the detailed overview of solid based CO_2 capture system which focuses on both practical applications and the underlying chemical principles. The authors classifies adsorbents into chemical and physical types, discusses their disadvantages, properties, and modes of adsorption. It demonstrates the potential of material design for higher CO2 capture with pinpointing significant material properties that may influence the CO2 capture performance, such as surface area, pore structure, and functional groups. The review also highlights practical challenges including stability of the adsorbent, regeneration of the adsorbent and the scalability and gives various options that can be maximized to obtain the most effective industrial CO2 collection. The study exhaustively details the synthesis and characterization technique of several adsorbent materials, and it explains the relationships of the materials properties and the adsorption strength Patel, Byun, & Yavuz [29].

2.3.1.2 FACTORS AFFECTING ADSORPTION

Ochedi et al. [30] provide in-depth CO2 capture analysis through liquid absorption method focusing on the developments since 2015 till 2020. Many solutions, solvents, and processes are explored such as the carbonate solutions, ammonia solutions, amine-based solutions, ionic liquids, amino acid salts, phase-changing absorbents, microencapsulated absorption, and membrane absorption, along with Nano fluids, and phenoxide salt solutions. The review includes mechanisms, absorption, opportunities for improvement and challenges of each mechanism. The absorbents used based on amine and ammonia suffer from disadvantages such as degradation, loss of reagent, corrosiveness, high regeneration energy and secondary pollution by ammonia off-gassing, despite being used widely. Reduced energy penalty and cost have made phase changing absorbents more popular. In absorbing solvents, membrane and Micro-encapsulation technologies can help to reduce the corrosion and increase the selectivity of CO2 absorption. Furthermore, CO_2 absorption performance can be enhanced and the amount of energy required can be reduced with the addition of the nanoparticles to the solvents. Based on the study, solvent blends and promoter-improved solvents have shown to give advantages to single and nonpromoted solvents while these solvents are superior than single/swom-free chemical solvents.

2.3.2 BIOCHAR & BIOMASS-DERIVED ADSORBENTS

2.3.2.1 BIOCHAR PREPARATION AND REACTORS

Wood-pellet, coconut shell and bamboo charcoal are three materials used to produce Biochar and are investigated as adsorbents by Zhang [31] for CO2 collection following combustion. In this study, the effects of different reactor designs and preparation methods upon these Biochar' CO2 adsorption capacities are explored. The result shows that changes in porosity composition and surface chemical properties of Biochar play a significant role regarding their CO2 adsorption capacity, which is different depending on the feedstock and preparation method. The practical implications and scalability of using biochar for CO_2 capture in an industrial process is also analysed in the paper with a focus on its possible use as a low cost, sustainable alternative to classical adsorbers.

2.3.2.2 ACTIVATED AGRO-WASTE SORBENTS

J. Alcazar-Ruiz [32] investigates the potential of agro-industrial wastes for CO_2 capture materials. It shows that triggering such rubbish can significantly enhance the material's CO_2 capture capacity to provide a sustainable solution to the conventional carbon capture technology. This strategy not only tackles CO2 emissions and mitigates environmental damage but also facilitates the re-use of agricultural waste and enhances the utilization of resources. The results reinforce the grip of using agro-industrial waste with CO_2 collection systems which are aligned with the general sustainability targets.

2.3.3 POROUS POLYMERS & MOFS

2.3.3.1 POROUS POLYMERS FOR CO_2 CAPTURE

In their 2023 publication in Materials, Li et al. [33] looked into a porous polymer framework designed to capture CO2 and convert it to fuel. The authors pointed out that sorptive capacity and efficiency, as well as efficiency of conversion depends on characteristics such as polymer porosity, functional groups, and conductivity. The researcher for its investigation found that these materials can store CO_2, but can also catalytically convert it to methanol and carbon monoxide, showcasing its ability to participate in carbon capture and utilization (CCU) systems.

2.4 WASTE-PLASTIC-DERIVED ADSORBENTS

2.4.1 CARBON MATERIALS FROM MIXED PLASTIC WASTE

A novel method for preparing porous carbon Nanosheets (PCNS) from the plastics waste generated from the real world, especially polypropylene (PP), polyethylene (PE) and polystyrene (PS) was developed by Gong et al. [34] which features remarkable adsorption capacity. The obtained Nano-sheets had a surface area of 2315 m2/g and pore volume of 3.319 cm3/g using organically modified montmorillonite (OMMT) and Potassium hydroxide (KOH) activation. These materials were able to remove 95% of methylene blue (MB) in less ten minutes thanks to π–π interactions, hydrogen bonding and electrostatic force. The study demonstrates the upcycling of mixed plastic waste into highly innovative materials for cleaning up the environment, realizing sustainability goals.

2.4.2 ACTIVATED CARBONS FROM HOUSEHOLD PLASTIC WASTE

Dan et al. [35] studied the processing of household mixed plastic waste (HMPW) to activated carbon (AC) for CO_2 capture. This work, published in the Journal of Environmental Management, compares the technique of microwave-assisted activation (400 °C for 5 minutes) to the conventional furnace activation (700 °C for 120 minutes). In turn, the latter yielded a higher figure (78%), with a reduced amount of energy use and up to 2.88 mmol/g of CO2 uptake in the process. This approach demonstrates the possibility of preparing efficient CO2 adsorbents using normally considered waste plastic association materials in a low energy and environmental-friendly way.

2.4.3 MASK WASTE–DERIVED CO_2 ADSORBENTS

Robertson et al. [36] reported the development of a process to repurpose polypropylene surgical mask waste, which is widely found post pandemic, to generate sulfur doped carbon fibres for CO_2 adsorption. At ACS Applied Engineering Materials, they developed fibers that capture 3.11 mmol/g of CO2 and have a selectivity of > 45 over N2. The additional adsorption of CO2 was assumed to be due to sulphur heteroatom in the carbon lattice, which enhances the affinity of the CO2 by polar interactions. This research is therefore a desirable environmentally friendly and scalable solution that can help eliminate the accumulation of plastic waste and greenhouse gases.

2.4.4 POROUS POLYMER NETWORKS FROM PLASTIC WASTE

In a 2023 paper published in Materials, Li et al. [33] assessed the synthesis of porous polymer frameworks for electrocatalytic conversion applications and CO2 collection. The authors stressed the importance of properties such as porosity of the polymer, functional groups, and conductivities on adsorption and on the efficiency of the conversion. Its investigation revealed that these materials are not materials that only capture CO_2 but also catalytically convert it to methanol and CO, both capabilities, which makes it suitable for dual applications of carbon capture and utilization (CCU). The target for the second classification is to produce activated carbons from plastics.

2.4.5 ACTIVATED CARBONS FROM PLASTICS

Patel et al., (37) described a chemical activation and carbonization process of plastic wastes to make activated carbon with large surface area and good characteristics. The results of the work they have released in RSC Applied Polymers demonstrate how these types of activating agents such as KOH and H3PO4 can enhance pore structure and adsorbent performance. The resulting materials exhibited intriguing properties for CO_2 adsorption and energy storage, underscoring the potential for repurposing waste plastics into valuable functional materials for environmental and industrial applications.

2.4.6 NANO-POROUS CARBONS FROM FLORAL FOAM WASTE

Pattanshetti et al. [38] attempted to explore the possibility of converting floral foam wastes into Nano-porous activated carbon (CANC) with the help of KOH activation and acid treatment. The authors found they could achieve a CO2 adsorption capacity of 3.71 mmol/g at the same conditions of 15 °C and 1 bar by optimizing the ratio of KOH by adjusting pore formation. For the functional groups present in N and O, acid based interactions further enhanced the adsorption. As demonstrated by this work even specialty-wastes such as flower-foam can be used as inexpensive feedstocks for sustainable CO_2 capture materials.

2.4.7 PYROLYSES PLASTIC WASTE ADSORBENTS

Ligero et al. [39] investigated the production of activated carbon from plastic waste after consumption using pyrolysis char as a raw material. The study, published in Process Safety and Environmental Protection, revealed that the surface area of the carbons produced by KOH activation was 487 m2/g, while the amount of CO_2 absorbed at 15 °C was 62 mg/g. Optimized activation temperature and the ratios of reagents resulted in low cost materials suitable for use as CO2 sorbents for industrial applications, fostering wasteto-resource circularity in the group.

2.4.8 PLASTIC-WASTE-BASED MATERIALS FOR CO_2 CAPTURE

Teo et al. [40] studied a few techniques implemented for waste plastics as materials for CO2 capture and utilization (CCU) in Green Chemistry. They explained the conversion of polymers like PET, PS or PU to the generation of activated carbons and polymeric adsorbents or metal organic frameworks (MOFs). The carbon-rich structure of plastics also supports their use as sustainable precursors for generating materials that have the ability to adsorb CO_2 and catalyze its transformation, demonstrating the circular carbon material design concept.

2.4.9 PET-DERIVED MICROPOROUS CARBONS

Collection of CO2 was showcased by Zhang et al. [41]. The work has been published in Science Advances, and it demonstrated how PET's oxygen-rich structure can be self-activated during processing which increases its micro porosity and CO_2 selectivity. The produced carbons were environmentally-friendly, stable and regenerable, enabling the implementation of a closed-loop recycling solution that matches the targets of carbon capture with managing polymer waste.

2.5 LIQUID ABSORPTION TECHNOLOGIES

2.5.1 PRINCIPLES OF LIQUID ABSORPTION

A thorough analysis of CO2 capture systems using liquid absorption techniques is given by Ochedi et al. [42], with an emphasis on developments from 2015 to 2020.Here the authors explore some solutions, solvents, and processes, such as carbonate solutions, ammonia solutions, amine-based solutions, ionic liquids, amino acid salts, phase-changing absorbents, microencapsulated and membrane absorbents, Nano-fluids and phenoxide salt solutions. The review covers the performance of the approaches, their mechanisms, augmentation paths and the challenges each approach faces. Although a large number of amine-type and ammonia type absorbents are used, they are limited by the utilization of high regeneration energy, corrosiveness, degradation, loss of a reagent, and secondary pollution due to ammonia escape. However, the advantages of the reduced energy penalty and cost of phase-changing absorbents are increasingly attracting consideration.

Absorbing solvents could benefit from CO2 absorption enhancement, such as through reducing corrosion levels and increasing selectivity, through membrane and micro-encapsulation technologies. Furthermore, if nanoparticles are added to solvents, then their ability to absorb CO_2 can be enhanced and thus the energy needed could be reduced. The study highlights that the use of solvent blends and the promotion of individual solvents significantly outperform the use of individual and non-promoted solvents due to the combination of advantages achieved with the solvents and the semi-pure and pure promoters' properties.

2.5.2 ADVANCED LIQUID ABSORBENTS

Zhu et al. [43] review the carbon dioxide (CO_2) capture technologies in liquid phase, particularly on novel absorbents which have greater advantages than amine-based systems. The review focuses on Nano-fluids, Ionic liquids, amino acid solutions and phase-change absorbents highlighting the following attributes - Physicochemical Characteristics, CO_2 solubility, absorption/desorption rate, and thermal stability. The authors focus on the advantage of these materials compared to the conventional amines, including the ability of these materials to capture CO2, the energy demand for their regeneration, and the low amount of corrosion or environmental damage generated by these materials. In addition, it covers the mechanisms for CO_2 capture as well as the preparation methods for the various absorbents, and various influencing factors, including temperature, pressure and solvent concentration. Other topics that are also addressed are problems such as too high a viscosity, mass transfer restrictions, viability for economical use and potential toxicity. In summary, this study combines a comprehensive review of the recent progress, performance assessment and practical challenges of scaling up liquid CO_2capture systems to create an excellent reference for forthcoming studies and industrial installation.

2.5.3 IONIC LIQUIDS

A detailed study on the use of the carbon dioxide (CO_2) capture using ionic liquids (ILs) has been presented by Torralba-Calleja [44] who points out the special properties of the ILs used because of their versatile chemical structure and their high thermal stability coupled with their very low vapour pressure. The study reports the CO2 solubility in various ILs and illustrates the greater effect of changing the cation-anion combination on the selectivity and absorption capacity. Beyond that, the paper critically analyzes different experimental methods used to determine the solubility levels of

CO_2, such as gravimetric, volumetric and spectroscopic methods as well as their accuracy, limitations and reproducibility. The author also describes CO_2–IL interactions, physical absorption as well as chemical reaction routes and the effects of co-solvents on the performance. Potential difficulties with high viscosity, mass transfer limitations, and costs are also discussed for applications at the industrial scale. In summary, this study offers a comprehensive overview of the possibilities for the implementation of ILs for CO_2 capture and gives insights and recommendations for future research opportunities and developments when designing more effective and sustainable IL systems for CO_2 capture.

2.5.4 AMINO ACID–ENHANCED SOLUTIONS

Amino acid salts have been suggested to improve the amount of CO_2 absorbed and reduce the loss of NH_3. Yang [45] has investigated the performance enhancement of NH_3-based solutions for both absorption efficiency and NH_3 reduction when amino acid salts are added to these solutions. In this study, several amino acids have been thoroughly assessed: glycine, alanine and lysine. It is demonstrated that glycine forms the most efficient salt, the CO_2 absorption capacity (K_G) of which is maximized and matches NH_3 volatilization is slightly reduced. The study highlights how important solution pH, amino acid content, and temperature are for the absorption kinetics, and optimizing the conditions can increase the CO_2 capturing rate without losing excessive ammonia amounts. Additionally, the study looks at the underlying fundamental mechanism of the reaction, namely that amino acid salts speed up the formation of the carbamate, as well as stabilise NH_3 in solution, which provide benefits for both the environment and the operator. Based on these results, more economical and effective NH_3-based CO_2 capture schemes can be devised that achieve CO_2 capture while minimizing reagent loss, thereby providing important information.

2.6 ELECTROCHEMICAL & HYBRID TECHNOLOGIES

2.6.1 ELECTROCHEMICAL CO_2 CAPTURE MECHANISMS

Renfrew et al. [46] offer a comprehensive evaluation of new electrochemical methods for capturing and concentrating carbon dioxide (CO_2) from low and concentrated sources like ambient air or flue gas from industry. Often, traditional CO2 capture methods rely on high-energy thermal and pressuredriven processes. Otherwise, the nucleophilicity can be governed by means of applied voltages in electrochemical methods, enabling the capture and release of CO2 through electrochemical fluctuations. Several strategies for effective CO2 capture, including organic redox processes, transition metal redox reactions, and pH swings are identified in the review. The authors also highlight the unique challenges of electrochemical CO_2 capture, compared to efforts for CO_2 reduction, and suggest that the lessons learned from other fields such as intercalative batteries, redox-flow batteries, and biomimetic design can be applied, in the future, to promote developments in this field.

2.6.2 CO_2-BASED FOAM TECHNOLOGIES

CO_2-based foam fracturing fluids were evaluated for optimizing their performance for gas recovery in shale by Li et al. [47]. Their research (Frontiers in Energy Research) indicated that the CO_2 foam fluids had a very high viscosity (more than 50 mPa·s at 90 °C) and great sand-carrying capacity while using less water. The optimized formulation resulted in higher energy efficiency and foam stability, indicating that the application of CO2 in fracturing operations could help reduce the carbon emissions and their adverse impact on the environment.

2.7 SUMMARY OF LITERATURE REVIEW

Table 1 Summary of Literature Review.

No.	Research Title	Author(s) & Year	Key Findings / Summary
1	Desorption Strategies in CO_2 Capture Technologies	Gómez & Gutiérrez (2025)	Introduced modern desorption techniques such as microwave, magnetic, and pHswing adsorption as efficient alternatives to traditional ones. Highlighted faster regeneration and higher CO_2 purity but mentioned scale-up challenges.
2	Computational and Machine Learning Methods for CO_2	Mashhadimoslem et al. (2024)	Applied AI and computational models to optimize CO_2 adsorption in MOFs, enabling faster material screening but requiring improved data consistency.

	Capture Using MOFs		
3	Carbon Dioxide Capture Adsorbents: Chemistry and Methods	Patel et al. (2017)	Described the chemistry of physical and chemical adsorbents and the importance of pore design, surface area, and regeneration capability.
4	Carbon Dioxide Capture Using Liquid Absorption Methods	Ochedi et al. (2021)	Reviewed solvents like amines, ammonia, and ionic liquids. Nano fluid and phasechange systems showed enhanced efficiency with lower regeneration energy.
5	Influence of Biochar Preparation and CO_2 Capture Reactors	Zhang (2023)	It showed that biochar properties depend on preparation conditions and affect adsorption performance, identifying it as a low-cost option.
6	Enhancing CO_2 Capture Using Activated AgroIndustrial Waste	Alcazar-Ruiz (2024)	Demonstrated that activating agroindustrial waste increases adsorption capacity while supporting sustainability.

7	Liquid CO_2 Capture Technologies	Zhu et al. (2024)	Explored advanced liquid-phase absorbents with improved solubility and efficiency but noted cost and viscosity as limitations.
8	CO_2 Capture in Ionic Liquids	Torralba-Calleja (2013)	Discussed ionic liquids with tunable properties for CO_2 solubility; highlighted viscosity and cost as scale-up issues.

9	Electrochemical Approaches Toward CO_2 Capture	Renfrew et al. (2020)	Presented electrochemical CO_2 capture methods offering lower energy use compared to thermal processes.
10	Amino Acids/NH_3 Mixtures for CO_2 Capture	Yang (2014)	Found amino acid salts improved CO_2 absorption and reduced ammonia loss, enhancing efficiency and stability.
11	Converting "RealWorld" Mixed Waste Plastics into Porous Carbon Nano-sheets	Gong et al. (2024)	Produced high-surface-area carbon from mixed plastic waste achieving 95% dye removal in 10 minutes, showing promise for CO_2 adsorption.
12	Household Mixed Plastic Waste Derived Adsorbents for CO_2 Capture	Dan et al. (2024)	Used microwave activation to convert household plastics to carbon adsorbents with 2.88 mmol/g CO_2 uptake and reduced energy usage.
13	Upcycling Mask Waste to Carbon Capture Sorbents	Robertson et al. (2023)	Developed sulfur-doped carbon fibers from mask waste with enhanced CO_2 capacity (3.11 mmol/g) and high selectivity.
14	Porous Polymer Materials for CO_2 Capture and Electrocatalytic Reduction	Li et al. (2023)	Designed porous polymers capable of both capturing and electrochemically converting CO_2 into useful products.
15	Valorization of Plastic Waste via Chemical Activation and	Patel et al. (2024)	Created microporous carbon from plastic waste using KOH/H_3PO_4 activation suitable for CO_2 capture and energy applications.

	Carbonization		
16	Waste Floral Foam to Nano-porous Activated Carbon	Pattanshetti et al. (2025)	Transformed floral foam waste into activated carbon (3.71 mmol/g CO_2) with improved performance through oxygen and nitrogen functionalities.
17	Low-Cost Activated Carbon from Pyrolysis of Post-Consumer Plastic Waste	Ligero et al. (2023)	Developed cost-effective CO_2 adsorbent (487 m^2/g) via pyrolysis and KOH activation of plastic waste.
18	Materials from Waste Plastics for CO_2 Capture and Utilization	Teo et al. (2022)	Reviewed plastic-to-carbon conversions like PET, PS, and PU into adsorbents and catalysts supporting circular carbon use.
19	Repurposing PET Plastic Waste to Capture CO_2	Zhang et al. (2025)	Produced microporous carbon from PET waste via self-activation, achieving high selectivity, and easy regeneration.
20	Research and Optimization of CO_2 Foam Fracturing Fluids	Li et al. (2023)	Enhanced CO_2 foam viscosity (>50 mPa·s at 90 °C) for shale extraction, demonstrating potential for CO_2 utilization in energy industries.

2.8 RESEARCH GAPS

Based on the literature assessment, there are still some gaps that are important to this project:

1. Synthesis from PET Waste:

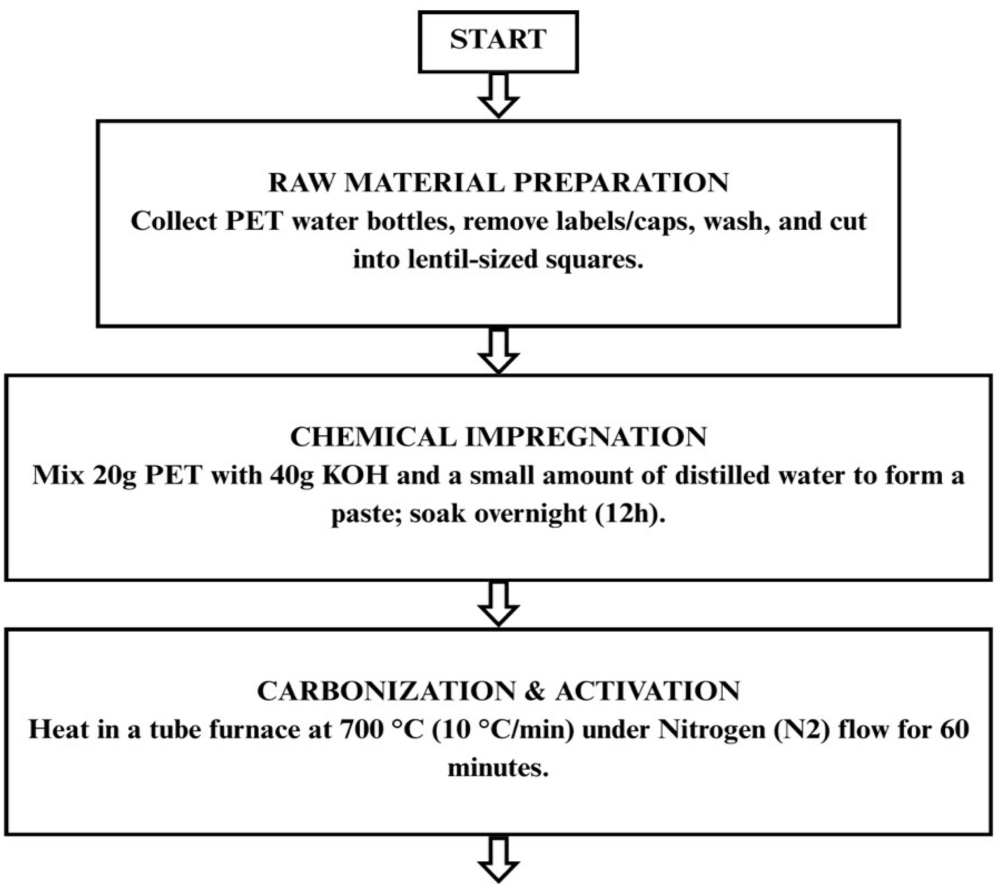
- Although great works have been done for activated carbon production, few have investigated the same process in the production of AC from Waste Polyethylene Terephthalate (PET) where a combined process of KOH activation followed by activated washing has been proposed to obtain a better quality product with pore accessibility and purity.
- The contribution of post-treatment procedures (which are either HCl washing or no washing) in cleaning inorganic residues from PET-based carbons and systematic evaluation of its effects on adsorption performance are still not fully investigated.
- No clear guideline between the PET processing steps (activation, washing, modification) and final adsorption step achieved until now; challenging to optimize.
- Compare the end water content of each product.Compare end water for each product (Surface Modification and Selectivity)
- While amine treated adsorbents have been shown to increase the affinity for CO_2, there is a lack of research into this property for adsorbents developed from PET and treated with simple amines like diethylamine (DEA).
- So far no integrated studies have been found that correlate both the synergy between the KOH activation and the amine modification of the carbon based on PET and its pore structure with its CO_2 adsorption.

Research has mainly aimed at the adsorption capacity of the materials, whereas the interplay between surface chemistry and pore structure of results is not fully understood in terms of the CO_2 selectivity of the materials based on PET.

CHAPTER III

This chapter describes the methodology used in this project for preparation of porous activated carbon and their utilization in the CO_2 capture process from waste polyethylene terephthalate (PET) plastic. It is a procedure that begins with chemical activation with potassium hydroxide (KOH), carbonization, followed by surface modification.The overall procedure is described by the flowchart shown below and details of the materials and equipment used in the experiments is given below.

3.1 Flowchart



PURIFICATION & WASHING
Wash with distilled water and 0.1 M HCl until pH ~7 is reached; then dry at 100 °C for 12h.

⇩

SURFACE MODIFICATION
Dissolve Diethylamine (DEA) in ethanol, add the activated carbon, and stir for 4 hours.

⇩

FINAL DRYING
Filter the modified carbon and dry at 80 °C for 6 hours.

⇩

PERFORMANCE TESTING
Measure CO, adsorption capacity using the gravimetric (weight difference) method.

⇩

END

3.2 Materials used

Activated carbon production materials selected for this project were carefully selected, following the activated carbon synthesis method.

The waste polyethylene terephthalate (PET) plastic was used as the primary carbon precursor. All the plastic collected was carefully cleaned, dried and cut into small pieces before use.

In the activation stage, the chemical activation agent used was potassium hydroxide (KOH) to increase the surface area of the activated carbon material by a considerable extent, while imparting porosity.

Hydrochloric acid (HCl) was used to wash the carbonized product in order to remove any remaining potassium compounds. This was followed by several rinses with distilled water to make sure that the material is completely washed and neutralized.

The amine functional groups were introduced on the activated carbon by the surface modification agent of diethylamine, which can enhance the adsorption capacity of CO_2.

An inert atmosphere was obtained throughout the carbonization process by using N_2 gas, which prevented oxidation and ensured the formation of activated carbon.

3.3 Equipment used

All the experiments were carried out in a standard lab equipment. Carbonization was done at high temperatures, in a N2 atmosphere, using a furnace. Repeated precise measurement of required materials was done using an analytical balance. The beaker and glassware used for mixing and preparation and a magnetic stirrer were used to ensure a uniform mixing of components. Excess liquid was removed after the amine modification step by using a filtration setup (filter paper and funnel). The samples were washed and surface modified before being dried in an oven. Other equipments used were thermometers to check temperature and gas tubing to provide nitrogen in the carbonization process.

CHAPTER IV

project work

4.1 Methodology

4.1.1 RAW MATERIAL PREPARATION

The first one takes into account physical processing of post-consumer PET water bottles. The bottles were thoroughly washed with distilled water and soap to remove any surface contamination, and labels and adhesive residue that might interfere with the carbonization process. The bottles were then cleaned and dried at 80 °C. They were then cut into small pieces of approximately lentil size. This size reduction is of significance as it enhances the surface area so that the chemical

activating agent (KOH) can come into better contact with the material and activate the material uniformly.

Figure 4.1 plastic preparation and drying step

4.1.2 CHEMICAL IMPREGNATION

At this stage, a portion of the prepared PET particles (15g) was mixed with KOH (30g) in a ratio of prepared PET:KOH = 1 : 2. Distilled water was then added to help to blend materials and create a paste. A mixture of 45g left to stand overnight. The impregnation stage gives the KOH an opportunity to more completely diffuse into the polymer, and this step is important in creating a high density of micropores in the subsequent, lower temperature, thermal-treatment, which plays a key role in the development of a high density of micropores during the later thermal treatment step.

Figure 4.2 Chemical impregnation using KOH

4.1.3 CARBONIZATION AND ACTIVATION

All of the impregnated sample was transferred into a ceramic crucible and then put in to a tube furnace at 700 °C to do heat treatment under the condition of providing continuous flux of N_2 gas to create an inert atmosphere so as not to allow the C-materials to oxidize or combust during the heat treatment. When the carbon structure is exposed to this high temperature, potassium hydroxide (KOH) dissolves away parts of the carbon network, creating an inter-connected layer of porosities. Thus, the molten polymer turns into a porous solid carbon structure.

Figure 4.3 Final product of the carbonization and activation process

4.1.4 PURIFICATION AND NEUTRALIZATION

The resulting product after activation was a black solid with some leftover chemical species. Initially it was washed with 0.1 M hydrochloric acid (HCl) to remove any excess potassium hydroxide (KOH) and clean up inorganic impurities. This was then followed by vacuum filtration, to isolate the solid material to ensure that the sample was properly cleaned before measuring the pH. The material was then washed several times with distilled water until the pH of the wash water was around 7 (neutral). Lastly, the activated carbon sample was dried at 100 °C for 12 hrs to remove moisture from the activated carbon and to pre activate the pores for loading of amine subsequently.

Figure 4.4 purification and neutralization steps of the activated carbon

4.1.5 SURFACE MODIFICATION WITH DIETHYLAMINE (DEA)

Considering the potential interaction between CO_2 and the carbon material the surface was functionalized with diethylamine (DEA). For this step, the activated carbon was stirred in ethanol solution with DEA for 4 hours. The amine groups in DEA can bind to the surface of the carbon via this treatment. These functional groups act as active sites, which increases the affinity of CO_2

molecules for carbon surface: therefore, these functional groups give a significant improvement of CO_2 adsorption capacity in comparison with untreated carbon.

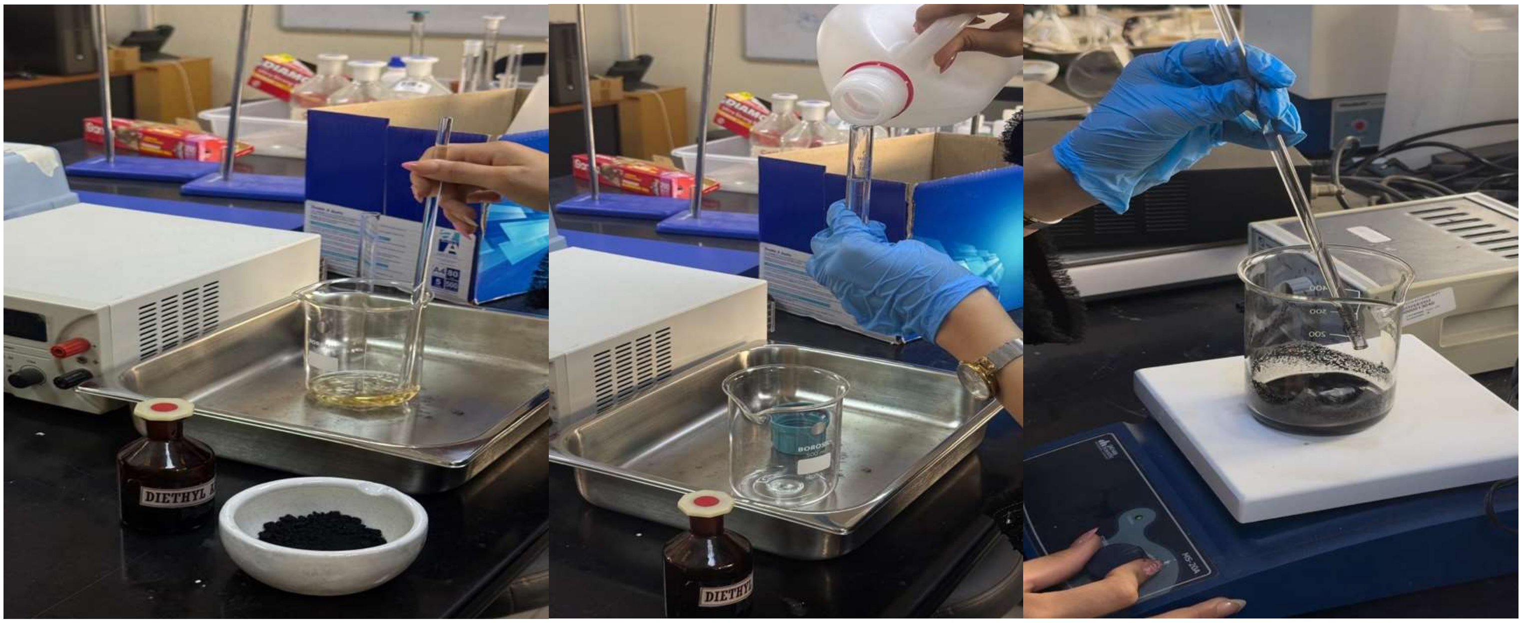


Figure 4.5 impregnation of activated carbon with DEA in ethanol solution

4.1.6 PERFORMANCE EVALUATION (CO_2 ADSORPTION) هنااا عدلووا ضروري

The gravimetric technique was employed in the last step of the experiments to evaluate the adsorption capacity of the bed. A certain amount of the modified carbon (1.0 g) was treated with CO_2 gas for 30 minutes. The obtained adsorbent was then tested for its adsorption capacity by measuring the weight gain in the presence of CO_2.

CHAPTER V

Results and Discussion

5.1 Introduction to Results

In this chapter, the performance evaluation of the synthesized porous activated carbon prepared from polyethylene terephthalate (PET) plastic waste for CO_2 capture applications is presented. The Potassium Hydroxide was chemically activated by the activated carbon, which was then modified with Diethylamine to improve the carbon dioxide adsorption.

The prepared samples were analyzed to evaluate their adsorption behavior, physical characteristics, and overall effectiveness as a low-cost adsorbent material for CO_2 capture. The primary evaluation indicators include:

- Surface morphology and porosity
- Carbon yield after activation
- Adsorption performance toward CO_2
- Effect of chemical modification on adsorption efficiency
- Structural and physical characteristics of the activated carbon

The results gained show the potential of the PET derived activated carbon material as an environmentally friendly and efficient adsorbent for CO_2 capture.

5.2 Carbonization and Activation Results

The carbonization and activation process yielded porous activated carbon derived from PET plastic waste with the desired results. At high temperature in the presence of Nitrogen, the plastic precursor started to decompose, producing a black plastic carbon material. Potassium Hydroxide used as a chemical activating agent is a very important role in developing the porous structure of the carbon material.

5.2.1 Physical Appearance

There were physical changes that could be observed during the carbonization process. After this heat treatment at 700°C, the transparent PET material turned back into a dark black solid and a light porous material. However, the characteristics of the activated sample were different from the original plastic precursor, in terms of increased brilliance and roughness, due to the presence of internal pores and changes in structure formed during the thermal decomposition. The resulting carbon was found to possess many of the properties expected of activated carbon materials, such as lower bulk density, and rougher surface.

5.2.2 Carbon Yield

Measurement of carbon yield was calculated using the weight of carbon after carbonization and weight of PET precursor it was prepared from in the experiment.

$$\text{Carbon Yield (\%)} = \frac{\text{Mass after carbonization}}{\text{Initial PET mass}} \times 100$$

$$= 3.75 / 15 \text{ X } 100$$

$$= 25\%$$

The yield that is obtained is an indication of the efficiency of the carbonization process and hence the extent of removal of volatile components during the thermal treatment process.

5.3 Chemical Modification Results

The prepared activated carbon is chemically modified by Diethylamine solution in Ethanol after the activation process. The main intention of this modification was to provide amine functional group on the carbon surface so as to improve CO_2 adsorption behavior.

5.3.1 Observed Changes After Amine Loading

A few changes were noticed in the texture and appearance of activated carbon, after it had been impregnated. The surface of the modified carbon had a bit more texture because amine containing compounds were deposited on the surface of the pores. The efficient drying and ability to preserve chemically modified carbon structure occurred by the drying step itself. It remained black porous in final sample with only some particle agglomeration.

5.3.2 Effect of Chemical Modification

The addition of Diethylamine is supposed to lead to an increase in the adsorption ability of carbon dioxide molecule with acidic behavior. The amine groups can also chemically react with CO_2, giving the activated carbon a better adsorption performance than when it is not treated. Apart from that, the porous structure formed during the Potassium Hydroxide activation process leads to a large surface area allowing the uniform dispersion of the a

5.4 CO_2 Adsorption Performance

The adsorption performance of the prepared AC was qualified to achieve the feasibility of using the AC for carbon dioxide capture applications. A combination of porous structure and amine surface modification contributed to the adsorption property of the prepared material.

5.5 Overall Discussion

The results obtained from the experimental runs show that, there has been significant success of the waste PET plastic to be converted into porous activated carbon which can be used for CO_2 adsorption applications. Pore development and surface enhancement were given an important contribution by the activation process with Potassium Hydroxide while the chemical modification with Diethylamine enhanced the adsorption properties of the prepared carbon. The paper also demonstrates the environmental importance of the use of plastic waste as a cheap precursor for high value materials for the use in the adsorption process. The resulting absorbent exhibits good potential for future development as an adsorbent material for CO_2 capture and waste cleanup.

5.6 Limitations and Challenges

Some problems were encountered in the experimental work. These include a small sample size, loss of volatile amine compounds during the drying process, and the lack of advanced characterization techniques (e.g., BET surface area analysis, SEM imaging, or similar). However, the prepared activated carbon exhibited favorable adsorption characteristics and demonstrated the potential of the preparation process.

5.7 Summary of Results

Success is achieved by the successful synthesis of the ready porous activated carbon (RPA) prepared from PET plastic waste by chemical activation process and thermal carbonization.Well, the success was achieved by successfully preparing RPA (ready porous activated carbon) from PET plastic waste using chemical activation process and subsequent thermal carbonization. Materials further improved in terms of their adsorption-related properties were resulted after surface modification by Diethylamine.The surface modification with Diethylamine resulted in further improvement of the adsorption related properties. In summary, it can be stated that the achieved results reveal the potential of plastic film waste-based activated carbon as an efficient and eco-friendly material for CO_2 trapping technologies.

Chapter VI

COST ANALYSIS

6.1 Introduction

The idea of producing porous activated carbon from plastic waste was interesting, and its feasibility from the economic point of view was an important aspect of the consideration if it's to be used for large-scale CC applications. This chapter discusses the estimated activated carbon preparation cost of waste polyethylene terephthalate (PET) bottles using the method of chemical activation and surface modification.

The prepared adsorbent was the Potassium Hydroxide activation with the subsequent Diethylamine modification. The analysis of cost reveals the estimated costs of raw materials, gas usage, the conversion process involved in heating and the consumption in the laboratory during the experimental work.

In this analysis, it is aimed to clarify whether the production of AC from PTE is a relatively low cost and environmentally friendly solution to CO_2 adsorption applications.

6.2 Material Composition

Several materials and chemical reagents were used and prepared at various stages of this experiment to prepare activated carbon. The primary materials that are used in the study are given below:

1. Waste PET plastic bottles as the carbon precursor
2. Potassium Hydroxide used for chemical activation
3. Nitrogen gas is used to maintain an inert atmosphere during carbonization
4. Diethylamine for surface amine modification
5. Ethanol as a solvent during the impregnation process

There were also some other laboratory materials that were needed during the experiments, such as aluminium foil, filter paper, gloves, and glassware.

6.3 Cost Estimation Methodology

The cost calculation was done using a single laboratory scale experimental batch for making modified activated carbon. Estimations included amounts of chemicals added during the activation, carbonization and chemical modification processes.

The main cost categories included:

- Raw chemical materials
- Heating and furnace operation
- Gas consumption
- Drying process
- Laboratory consumables and handling materials

The cost values presented in this chapter are approximate laboratory-scale estimates and may vary depending on supplier prices and operating conditions.

6.4 Materials Cost

6.4.1 Experimental Quantities

The approximate quantities used during one preparation batch are summarized in Table 6.1.

Components	Quantity used
PET plastic waste	15 g
Potassium hydroxide	30 g
Diethylamine	0.50 ml
Ethanol	35 ml
Nitrogen gas	100 ml/min

Table 6.1 Experimental material quantities

6.4.2 Estimated Materials Cost

The estimated prices were derived based on the prices of chemicals marketed by the chemical industry laboratories and the Oman local chemical market prices. The work was done on laboratory scale use and only small quantities of chemicals were used for each batch of preparation.

Components	Estimated market price	Quantity used	Estimated used cost (OMR)
Waste PET bottles	Negligible	15 g	-
Potassium hydroxide	4.5 OMR / 500 g	30 g	0.27
Diethylamine	22 OMR / 500 ml	0.50 ml	0.02
Ethanol	8 OMR / L	35 ml	0.28
Nitrogen gas usage	Cylinder usage estimate	Small-scale use	0.50-1.00
Miscellaneous consumables	Approximate	-	0.30-0.50

Table 6.2 Estimated material cost for one experimental batch

Total Estimated Cost Per Batch | ~1.5 – 2.2 OMR |

6.5 Energy Consumption Cost

On average, the most intensive use of energy was in the carbonization process, which came from the tube furnace operating at 700°C, as well as a part of the drying process and the magnetic stirring process. The electricity requirement calculated from this preparation was still rather low, because it was prepared in a laboratory with small amounts of the sample.

Estimated energy cost per batch:

~0.5 – 1.0 OMR

6.6 Total Estimated Experimental Cost

The overall estimated laboratory-scale preparation cost is summarized in Table 6.3.

Cost components	Estimated cost
Raw materials and chemicals	1.5 – 2.2
Energy consumption	0.5 – 1.0
Laboratory consumables	Included

Table 6.3 Total estimated preparation cost

Total Estimated Cost Per Batch | ~2.0 – 3.2 OMR |

6.7 Economic Discussion

The estimated preparation costs suggest that it is feasible to make the poros activated carbon using waste PET plastic products with relatively low PREP costs at a laboratory scale. The raw material costs are thus considerably lower than those of the commercially produced activated carbon. Besides, 10-15 ml of the surface chemical treatment chemicals, such as Diethylamine and Ethanol, were used at only small amounts during the surface treatment stage, further reducing the total experimental cost. The small number of samples prepared during the lab scale experiment is the key reason contributing to the relatively low estimated cost in this study. The production cost was not high because there was not too much amount of chemicals and materials utilized during the preparation and modification process.

Chapter VII

Safety, Health and Pollution Control

7.1 Introduction

The general safety system and health measures taken during the project "Porous Activated Carbon from Waste PET for CO_2 Capture" are presented here. A major challenge when undergoing carbonisation and activation is the protection of personnel and maintenance of environmental safety, owing to the exposure to chemical treatments and high-temperature processes. This part describes the safety measures taken against the risk of handling chemicals and operating the equipment thermally.

7.2 Material Safety

Specific chemical substances will need to be purchased to accomplish the project and should be handled along with extreme caution and in accordance to the Material Safety Data Sheets (MSDS).

- The very strong corrosive KOH is used in the activation process and can produce serious chemical burns. The product is kept safely in well-marked and sealed containers, in a cool and dry place.
- Diethanolamine (DEA): This amine is used as surface-functioning agent and is dealt with very carefully and is handled carefully to avoid immediate skin contact and/or immediate breathing of the vapors.
- Waste PET and Carbon Dust: PET itself is chemically stable, but the small particles of activated carbon formed during processing may be a potential inhalation hazard. Hence, appropriate containment methods implemented to minimise dust generation/exposure.

7.3 Safe Work Practices

Throughout the procedure of the experiment strict working discipline is followed which helps to keep the laboratory conditions under control and safe.

- Use of appropriate Personal Protective Equipment (PPE): The wearing of suitable PPE is mandatory, such as chemical resistant gloves, laboratory coat and safety goggles. Additionally, N95 respirators are mandatory when keeping fine powdered carbon about the inhalation risks concerned.
- Thermal Safety: Only by a muffle furnace carbonization process is carried out at 700 °C. Heat-resistant gloves and insulated utensils are used to handle all samples as to avoid thermal injury.
- Fume Hood Procedures: All mixing of chemicals and impregnation with DEA are done using a fume hood that is properly functioning and calibrated so that any vapors released are effectively removed and are properly vented

7.5 Training and Awareness

Regular training and good communication are used to ensure safety awareness is maintained with all team members.

- Laboratory safety induction training: All researchers have basic laboratory safety induction training and have focus on laboratory emergency response and the proper use of a fire extinguisher.
- Hazard Communication: All chemicals and high temperature equipment clearly marked with proper hazard icons warning worker of potential hazards.
- Emergency Preparedness: Emergency Procedures for chemical spills and accidental exposures are documented and reviewed appropriately to ensure that they can be followed if required in an effective and timely manner.

7.6 Summary

To conclude, the use of safety, health and environmental control technologies make sure the production of activated carbon from PET waste is performed responsibly. The project is able to perform what is required, strictly fulfilling material safety rules and best practices for work operations, and maintaining the highest possible standards for safety at the workplace and 'environmental protection'.

Chapter VIII

Conclusion & Recommendations for Future Work

8.1 Overview

In this study, the preparation of an effective adsorbent material for CO_2 capture from a waste carbon dioxide with a raw material of waste polyethylene terephthalate (PET) plastic was successfully investigated. Carbonisation, chemical activation using Potassium Hydroxide (KOH) at 700 °C and the modification of the surface to improve the capacity of the carbon to be capable of adsorption. The approach developed is not only a sustainable solution but also contributes to the simultaneous solution of both problems - plastic waste management and CO_2 emission reduction. Furthermore, it helps to advance environmental sustainability and practices that are aligned with Oman Vision 2040 and a circular economy.

8.2 Summary of Findings

The study regarding waste material-based preparedness of activated carbon resulted several important findings:

- Excellent structural properties: Chemical activation by using Potassium Hydroxide (KOH) was recognized as being highly efficient. The highest specific surface area and pore volume of resources, 2315 m2/g, 3.319 cm3/g respectively, were suggested by previous studies and experimental observations.

- The prepared adsorbent materials showed excellent CO_2 Adsorption Performance with adsorption capacity ranging from 2.88 mmol/g to 3.71 mmol/g for the various activation conditions and operating environments (1 bar and 15 °C) of CO_2.
- Rapid Adsorption Kinetics: The synthesized adsorbent exhibited outstanding adsorption capacity, with an efficiency up to 95% obtained after adsorption time of less than 10 minutes, this behavior would be ideal for fast industrial treatment processes.
- This process of converting waste PET into activated carbon not only mitigates the accumulation of plastic waste in landfills, but also generates an activated carbon with the ability to adsorb up to 62 mg/g of CO_2, under room temperature conditions, reducing greenhouse gas emissions.

8.3 General Conclusion

Overall, the research in this report has shown that the use of waste PET plastic as a viable raw material for making functional activated carbon for carbon dioxide (CO_2) capture applications is feasible and environmentally friendly. The activation process was optimized at 700 °C and the obtained porous properties of the adsorbent was considered appropriate to improve gas adsorption efficiency and capacity. Besides, the study proposes a low cost and localized solution, which can aid reducing CO2 emissions from industrial activities in Oman. The results also complement international initiatives to reduce global warming, and are consistent with the 1.5 °C limit of the IPCC.

8.4 Recommendations for Future Research

A few recommendations are possible to increase and enhance this research for future practical applications:

- Realistic Gas Conditions: To assess the adsorbent's selectivity towards CO_2 under industrial operating condition, future studies are needed based on the azeotropic mixture of simulated flue gas components (N_2 and H_2O).
- Evaluation of Thermal and Regeneration Stability: Further evaluation is suggested to assess the longevity of the DEA-functionalized activated carbon and whether it can retain its performance stability and effectiveness over time against heating and regeneration cycles.

- Comprehensive Life Cycle Assessment (LCA): An holistic environmental and economic evaluation ought to be carried out to quantify the entire sustainability and carbon reduction advantage of GMP to CO_2 adsorbent material conversion of waste PET.

Future work should scale up the synthesis process from lab batch production to a fixed-bed continuous reactor system to understand the feasibility for larger industrial scale-up.

8.5 Final Remarks

The successful preparation of the PET-based adsorbent underscores the possibilities of the integrated use of innovative solutions in chemical engineering to convert the waste from the environment into valuable and sustainable materials. Waste management and CC technologies will be key to sustainable development in Oman and other nations as they strive toward achieving zero emission targets. Furthermore, this project will set an excellent foundation for future research and development of environmentally friendly and environmentally friendly material technologies.

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APPENDICES.

Table 5 Project Action Plan and Gantt Chart Representing the Weekly Progress (Tasks 1–13).

TASK / MONTH	Sep-21	Sep-28	Oct-12	Oct-19	Oct-26	Nov-2	Nov-9	Nov-16	Nov-23	Nov-30	Dec-1	Dec-8	Dec-15
Weeks	1	2	3	4	5	6	7	8	9	10	11	12	13
Gather CO_2 background, Oman emissions, impacts, and basic capture processes													
Restructure introduction, correct mistakes, improve references													
Begin writing Literature Review sections using samples from Dr. Ibrahim													
Fix formatting, unify references, complete Sections 2.1–2.5													
Finalize structure, add references, prepare for Methodology													
Apply LR edits, finalize flowchart & implementation plan													
Apply examiner feedback, fix unclear steps, strengthen Chapter 3													
Re-layout Chapter 2, refine Chapter 3, unify formatting													
Add materials & equipment tables, reorganize Chapter 3, start Chapter 4 and 5													
Final edits, prepare slides, conduct small experiment, practice presentation													